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Tethysian Metallogenesis of the Major Metalliferous Minerals in Mainland SE Asia and Sanjiang Region in SW China: Implications for Exploration

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The Sanjiang region in Yunnan (southwestern China) and its adjacent mainland SE Asia comprises a complex collage of microcontinents including the Indochina, Sibumasu and West Myanmar Terranes(Zhong,1998; Metcalfe,2002). Within these microcontinents,there are a number of suture zones that represent the remnants of the three Tethyan Oceans (Paleo–Tethys, Meso–Tethys and Ceno–Tethys) and their branches or oceanic backarc basins that once existed in this region(Li et al.,2004; Metcalfe,2013).

1 Deposit Types and Metallogenic Epochs

By summarizing and correlating the characteristics of ore deposit types in the Sanjiang and mainland SE Asia regions(Hou et al.,2007;Khin Zaw et al.,2007,2013;Deng et al.,2013a,2013b;Wang et al.,2013), eleven major types of ore deposits (e.g., VHMS Cu-Pb-Zn-Ag,SEDEX Pb-Zn-Cu -Ag,basin–related Pb-Zn-Cu-Ag-Co, porphyry –related skarn Pb-Zn-Cu-Ag-Sn-Fe, porphyry/porphyry –related skarn Cu-Au(Mo), epithermal Au-Ag,sediment –hosted/ orogenic Au, sediment–hosted/ Carlin–like Au -As-Sb-Hg,cassiterite–quatz vein/greisen/pegmatite Sn- (W), magmatic sulfide Cu-Ni and hot spring Au) and eight mineralization epochs (e.g.,Early Paleozoic, Middle Devonian–Early Carboniferous, Late Carboniferous– Early Permian, Late Permian–Middle Triassic, Late Triassic-Early Jurassic, Late Jurassic-Early Cretaceous, Late Cretaceous-Paleocene and Eocene–Miocene) have been identified.

2 Regional Tectono-Metallogenic Correlation

We suggest that the opening and closure of the Tethyan oceans and the collision between the microcontinents have played a major control to the metallogenic relation in the region(Hou et al.,2007;Khin Zaw et al.,2007, 2013;Deng et al.,2013a,2013b),.During the evolution of the Prototethyan margin (Ordovician to Silurian) and Paleo–Tethys (Devonian to Triassic), continental rifting and generation of ocean basins together with the Early Paleozoic opening phase have led to the formation of SEDEX Pb-Zn-Cu-Ag deposits (e.g., Silurian Mengnuo Pb-Zn deposit, Yunnan) in continental passive margin, as well as VHMS Cu-Pb-Zn-Ag deposits (e.g., Silurian Dapingzhang deposit, Yunnan; Cambro–Ordovician Bawdwin deposit, Myanmar) in ocean basin/supra– subduction backarc basin spreading centres (Lehmann,et al.,2013).

In the Late Paleozoic–Early Mesozoic closure phase, ore deposits are mainly represented by porphyry–related skarn/ epithermal Cu-Au deposits (e.g., Late Permian Chatree epithermal Au-Ag deposit, Thailand) along an Andean–type continental magmatic arc. This is followed by syn–collisional sediment–hosted/orogenic gold deposits along the sutures (e.g., Triassic Selinsing Au deposit, Malaysia), together with late– to post–collisional granitoid–related Sn-(W) deposits (e.g., Late Triassic–Early Jurassic Shiganghe deposit, Yunnan). Among these Prototethyan and Paleotethyan deposits, we suggest that the porphyry–related skarn/epithermal Cu-Au (-Mo) deposits and sediment–hosted/orogenic Au deposits are the most important exploration targets.

The Early Cretaceous porphyry–related skarn Pb-Zn- Cu -Ag-Sn-Fe deposits (e.g., Late Cretaceous Shizhishan deposit, Yunnan) and Late Cretaceous–Paleocene granitoid –related Sn-(W) deposits (e.g., Xiaolongshan and Lailishan deposits, Yunnan) are the main deposit types associated with the evolution of Meso–Tethys. Late Cretaceous-Paleocene granitoids are associated with the Sn–mineralized belt in mainland SE Asia which extends over

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several 1000km (from Tengchong Yunnan, through central Myanmar, west Thailand and west of central Mountain Ranges of Malaysia, to Sungai Rokan Sumatra).The types of deposits associated with Ceno–Tethys are mainly high sufidation epithermal Cu - (Au) deposits (e.g., Miocene Monywa deposit, Myanmar) and epithermal/ sediment–hosted Au-Ag deposits (e.g., Kyaukpahto deposit, Myanmar).

Cenozoic basin-related Pb-Zn-Cu-Ag-Co (e.g., Paleogene Jinding deposit, Yunnan) are mainly hosted in Meso–Cenozoic mottled clastic rocks and strictly controlled by the Cenozoic thrust systems(Hou et al.,2008). Eocene porphyry/porphyry–related skarn Cu-Mo-Au deposits (e.g., Eocene Baiya deposit, Yunnan） occur in the non–arc settings, and related to melting of thickened mafic lower crust. These two types of deposits have significant exploration potential in Sanjiang and mainland SE Asia. Their metallogensis were associated with the collision between Indio and Eurasia continents, which correspond to the main– and late–collisional stages of the Indo–Asian orogeny.

3 Implications for Exploration

Interestingly, although the Sanjiang Region and the neighbouring mainland SE Asia show a similar geologic and metallogenic setting, mineral deposit discoveries in these two regions have certain differences. For instance, world–class basin–related Pb-Zn and Cu - Ag -Co deposits of Paleogene age have been discovered in the Langping–Simao basin, but large deposits of these types have not yet found in its southern continuation in the Indochina and Sibumasu terranes. In addition, economically important Permian–Triassic porphyry– related skarn/epithermal Cu-Au-Mo deposits have been widely documented along the Truong Son and Loei fold belts in the Indochina Terrane, yet similar deposits remain largely unknown in their northern extension along the western Ailaoshan Fold Belt in SW Yunnan. The distribution of ore deposits above suggests an enormous mineral potential in both the Sanjiang and mainland SE Asia regions, and metallogenic research on these two regions will be critical in future mineral exploration and discovery.

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References

Bernd Lehmann, Zhao Xinfu, Zhou Meifu, Du Andao, Mao Jingwen, Zeng Pusheng,Friedhelm Henjes-Kunst and Klaus Heppe,2013 Mid-Silurian back-arc spreading at the northeastern margin of Gondwana:The Dapingzhang dacite-hosted massive sulfide deposit,Lancangjiang zone, southwestern Yunnan, ChinaGondwana Research, 24: 648–663

Deng Jun, Wang Qingfei,Li Gongjian.,Li Chusi and Wang Changming,2013a.ethys tectonic evolution and its bearing on the distribution of important mineral deposits in the Sanjiang region, SW China.Gondwana Research, http://dx.doi.org/ 10.1016/j.gr.2013.08.002.

Deng Jun., Ge Liangsheng., Yang Liqiang, 2013b.Tectonic dynamic system and compound orogeny:additionally discussing the temporal-spatial evolution of Sanjiang orogeny, Southwest China. Acta Petrologica Sinica,29, 1099–1114 (in Chinese with English abstract).

Hou Zengqiang., Khin Zaw., Pan Guitang, Mo Xuanxue, Xu Qiang, Hu Yunzhong., Li Xingzhen, 2007. Sanjiang Tethyan metallogenesis in S.W. China: tectonic setting, metallogenic epochs and deposit types. Ore Geology Reviews 31, 48–87.

Hou Zengqian, Song Yucai, Li Zheng, Wang Zhaolin, Yang Zhiming, Yang Zhusen, Liu Yingchao, Tian Shihong, He Longqing, Chen Kaixu, Wang Fuchun,Zhao Chengxiang, Xue Wanwen and Lu Haifeng,2008.Thrust-controlled, sediments-hosted Pb–Zn–Ag–Cu deposits in the eastern and northern margins of Tibetan orogenic belt: geological features and tectonic model. Mineral Deposits 27, 123–144 (in Chinese with English abstract)

Li Xingzhen, Liu Chaoji, Ding Jun,2004.Correlation and connection of the main suture zones in the Greater Mekong subregion. Sedimentar y Geology and Tethyan Geolog y, 24 (4) : 1-12 (in Chinese with English abstract).

Metcalfe I.,2002.Permian tecto nic framework and palaeogeography of SE Asia.Journal of Asian Earth Sciences, 20: 551-56.6

Metcalfe, I., 2013. Gondwana dispersion and Asian accretion: Tectonic and palaeogeographic evolution of eastern Tethys. Journal of Asian Earth Sciences,66,1-33.

Wang Changming,,Deng Jun,Emmanuel John M. Carranza, M. Santosh,2013.Tin metallogenesis associated with granitoids in the southwestern Sanjiang Tethyan Domain: Nature,deposit types, and tectonic setting, Gondwana Research , http:// dx.doi.org/ 10.1016/j.gr.2013.05.005.

Zaw Khin, Meffre Sebastien, Lai Chun-Kit, Santosh, M., Burrett Clive, Graham Ian, Manaka Takayuki, Salam Abhisit, Kamvong Teera and Cromie Paul, 2013.Tectonics and metallogeny of mainland Southeast Asia – A review and contribution.Gondwana Research, doi: 10.1016/j.gr.2013. 10.010.

Zhong Dalai,1998.Paleotethysides in west Yunnan and Sichuan, China. Beijing: Science Press, 231 pp. (in Chinese).

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